FlexiMetal™: A Pump Out Resistant High Performance Metallic Gel-Like Thermal Interface Material for CPU, GPU and Other Electronic Cooling Applications

Himanshu Pokharna[1*], Sangram Kumar Samal[2], Yogesh Fulpagare[3], Chi-Chuan Wang[2]
[1]Deep Materials Inc, Saratoga, CA USA
[2]Department of Mechanical Engineering, National Yang Ming Chiao Tung University, Hsinchu City, Taiwan
[3]Department of Mechanical and Aerospace Engineering, The University of Texas at Arlington, Texas, USA
Corresponding email: pokharna@deep-materials.com

***Abstract*—** Increasing power densities in datacom, artificial intelligence (AI) accelerators, graphics processing units (GPUs), and high-performance client systems demand thermal interface materials (TIMs) that combine low thermal resistance with long-term mechanical and chemical stability. Conventional polymer-based greases and phase change materials (PCMs) are constrained by limited intrinsic conductivity and by degradation mechanisms such as pump-out, dry-out, and void formation, while liquid metal TIMs introduce challenges in handling, containment, corrosion, and electrical safety. This paper presents FlexiMetal™, a metallic, gel-like TIM engineered to deliver solder-like through-thickness thermal performance with grease-like compliance and non-pumpable stability. FlexiMetal employs a carbonaceous conductive particle network embedded in a metallic matrix. Its elastic, solid-like response arises from the packing and ordering of the metallic and carbonaceous phases; the material is predominantly elastic at rest yet shear-responsive during assembly and dynamic die movement. We describe the material design principles, steady-state performance in a dedicated thermal test vehicle (TTV) and a notebook platform, comprehensive evaluation in direct-to-chip liquid cooling systems, and an extensive reliability test matrix including thermal shock, power cycling, high-temperature storage, and high-temperature/high-humidity exposures. Across these evaluations, FlexiMetal exhibits low and repeatable interface resistances with no significant directional drift, indicating suitability for large dies, multi-die modules, and cold-plate-cooled high-power systems.

Keywords—Thermal interface materials, pump out, Liquid cooling TIM

I. INTRODUCTION

Thermal interface materials (TIMs) have become a central design element in electronic cooling because real mating surfaces—dies, lids, heat spreaders, vapor chambers, and cold plates—are never perfectly flat or smooth. Even nominally polished surfaces contact only at discrete asperities, leaving micro- and macro-scale gaps that, if unfilled or poorly filled, contribute substantial thermal contact resistance in series with the package and cooling solution. As device power, heat flux, and package complexity have increased, the allowable junction-to-ambient budget has tightened to the point where interface resistance can dominate overall thermal performance. Prasher's seminal review formalized this problem by decomposing the thermal interface into bulk conduction through the TIM, the bond line thickness (BLT), and the contact resistances at each surface, and showed that practical TIMs must (i) form continuous conductive paths across asperities, (ii) achieve thin and uniform BLTs under realistic clamp loads, and (iii) maintain these properties over time under thermal cycling, warpage, and material aging [1].

Conventional polymer-based greases and phase change materials (PCMs) have enabled several generations of products but exhibit inherent limitations against these requirements. Their effective thermal resistance is constrained by modest intrinsic conductivity and by degradation mechanisms such as pump-out, bleed, dry-out, and filler redistribution, which are exacerbated in long-life or high-cycle use. Pump-out (material transport out of the bond line) and dry-out (phase separation of particles and polymer) have been extensively documented as dominant degradation modes in commercial thermal greases subjected to thermal and power cycling. Recent work by Nagrani and Marconnet systematically correlates void formation, thermal resistance growth, and rheology in multiple commercial greases under combined thermal and mechanical cycling, and highlights that even high-performance formulations remain susceptible to pump-out and voiding under realistic thermo-mechanical loads [2].

To push performance further, some high-end systems have adopted conventional gallium-based liquid metal TIMs between the processor and heatsink. These low-viscosity, electrically conductive alloys can deliver very low initial BLT and excellent thermal resistance; however, they introduce significant integration and reliability challenges:

1. The liquid, highly wetting, conductive alloy must be fully contained by a robust, effectively "liquid-tight" barrier system to prevent leakage onto adjacent components. These barrier systems have been well adopted in laptops, but are expensive and very hard to rework, especially in the field. This severely limits adoption.

2. The extremely low viscosity and mobility make the material susceptible to pump-out or redistribution under thermal and mechanical cycling, especially in long-term field conditions.

3. Manufacturing and integration are limited: such TIMs generally cannot be pre-applied on a heatsink or cold plate and shipped as an integrated subassembly using standard processes, complicating assembly flows and further complicating field service.

Low-melting solders and other rigid metallic TIMs address some containment concerns but at the cost of higher mechanical stress, challenging rework, and increased risk for large or fragile packages. These trade-offs become more severe as packages evolve toward large dies, multi-die and chiplet-based modules, stacked memory, and direct cold-plate cooling, where non-ideal flatness, global and local bow, and dynamically changing curvature (convex/concave/flat) create strong spatial variation in gap thickness that traditional greases, PCMs, solders, and free-flowing liquid metals all struggle to accommodate reliably and repeatably.

Modern high-power systems therefore require TIM solutions that go beyond simply increasing filler loading or nominal conductivity. An effective next-generation TIM must combine (i) high intrinsic through-thickness conductivity, (ii) rheology and yield behavior that allow it to conform to rough, bowed, and warped surfaces at safe mechanical loads, (iii) geometric and chemical stability that suppresses pump-out, leakage, and uncontrolled migration, and (iv) robustness to thermal shock, humidity, and power cycling without directional drift in performance. These needs motivate structured, gel-like, and metallic composite TIMs that are explicitly engineered to couple conduction, wetting, and shear-responsive, elastic mechanical behavior to the realities of large-area, non-planar, and dynamically loaded interfaces [4].

FlexiMetal has been developed within this context as a metallic, gel-like thermal interface material intended to capture much of the thermal performance associated with conventional gallium-based liquid metal TIMs, while offering handling, containment, assembly, and potential rework characteristics closer to high-quality greases or pastes. Its architecture is designed to provide metal-rich, low-resistance thermal pathways together with yield-stress, viscoelastic behavior that confines the material to the bond line, resists pump-out, and supports manufacturable integration into high-power client, data center, and AI platforms. The following sections describe its material design principles, measured performance in a controlled thermal test vehicle and notebook platform, comprehensive evaluation in modern liquid-cooling systems, and results from an extensive reliability test matrix evaluating its suitability for large dies, multi-die modules, and cold-plate-cooled systems.

II. Material Design Principles

FlexiMetal is formulated as a metal-rich composite in which a continuous liquid-metallic phase is structured by a high-conductivity particulate network based on an advanced carbon allotrope. This architecture provides robust through-thickness thermal pathways while enabling the bulk material to behave as a structured metallic gel. At operating conditions and small strains, FlexiMetal exhibits a predominantly elastic, G' (storage modulus)-dominated response that arises from the crowding, jamming, and ordering of the carbonaceous particle network within the liquid-metal matrix. This solid-like network limits uncontrolled flow and pump-out, maintains geometric confinement within the intended bond line, and helps preserve a stable, well-defined interface over time. In this work, we use the term "viscoelastic" only in a generalized, macroscopic sense; the time-dependent response is governed by the particulate network in the metallic matrix rather than by a continuous organic polymer phase.

Under elevated shear during assembly, the material exhibits reversible shear-thinning and thixotropic behavior. This enables improved wetting of surface asperities, displacement of trapped air, and mitigation of interfacial voids. The formulation is compatible with copper and nickel-plated copper surfaces, promoting intimate metallic contact and low interfacial thermal resistance without relying on a rigid solder joint. The gel-like structure and yield stress confine the TIM to the intended bond line region, reducing the risk of migration into adjacent components compared with unconstrained liquid metals.

III. Experimental Methodology

A dedicated thermal test vehicle (TTV) was constructed to compare FlexiMetal against conventional greases and PCMs under controlled conditions. The TTV comprises a copper heat source block with a nominal contact area of 1.5 cm × 1.5 cm (2.25 cm²), cartridge heaters embedded in the block, thermocouples monitoring heat source and ambient temperatures, and a common heatsink and fan assembly shared across all materials. Clamping force was applied using a four-screw fixture referenced to mechanical stops. Pressure mapping film and mechanical calculations were used to target an average interface pressure of approximately 30 psi.

Complementary evaluations were conducted in a commercial notebook platform using a high-performance mobile processor and a production thermal module. Multiple reference TIMs and FlexiMetal were tested under identical conditions using a processor power stress tool at fixed power while logging junction temperature and system telemetry. Power cycling was used to simulate die dynamic movement to study potential pump-out issues.

Lastly, to assess FlexiMetal performance in modern liquid-cooling architectures, FlexiMetal was evaluated alongside a high-end grease-type reference material in a direct-to-chip cold plate system. This evaluation employed a Distribute Inlet Outlet Jet Impingement Cooling Cold Plate (DIOJIC-CP), manufactured using electrochemical additive manufacturing (ECAM) 3D printing with copper, mounted on a dedicated TTV with a 30 × 30 mm² heated copper block driven by 1000 W of dissipated power [5, 6]

Two thermal interface materials were compared in this system:

- **Grease TIM**: Cooler Master CryoFuze7, declared thermal conductivity $k \approx 14$ W/m·K
- **Soft metal TIM**: FlexiMetal, measured thermal conductivity $k \approx 35$ W/m·K

Both materials were applied at a nominal initial thickness of 0.2 mm over the heated surface. The coolant medium was a propylene glycol–water mixture (PG25), circulated at inlet temperature 45 °C and flow rates spanning 0.75–2.5 L/min. Five thermocouples were strategically positioned on the heated surface to simultaneously capture center, average, and maximum surface temperatures, enabling comprehensive thermal characterization. Pressure drop across the cold plate and coolant inlet/outlet temperatures were recorded with data acquisition at steady-state conditions (temperature and pressure stable within ±0.2 °C and ±1%, respectively). This setup enabled direct quantification of the thermal benefit of FlexiMetal over conventional greases in next-generation high-power cooling systems.

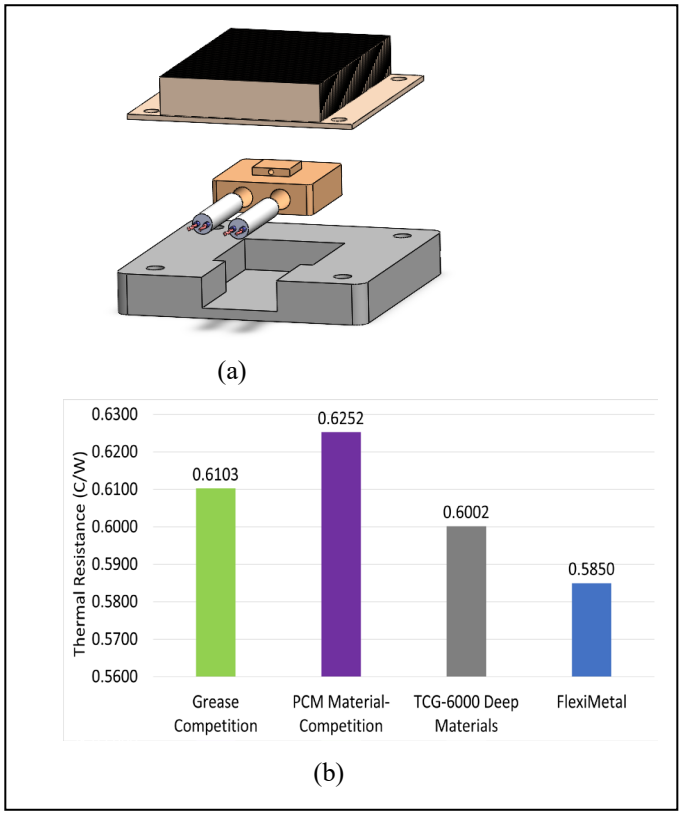


Fig. 1. 25 cm² Thermal test vehicle (TTV) configuration used to compare FlexiMetal and reference TIMs and (b) Test data

IV. Performance Test Results

In the TTV configuration (Fig. 1a), FlexiMetal achieved significantly lower thermal resistance than leading polymeric greases and PCMs evaluated under the same geometry and load. For the 1.5 cm × 1.5 cm contact area, measured interface resistances below approximately 0.02 °C·cm²/W were obtained at practical mounting pressures. As can be seen in Fig. 1b, conventional organic materials exhibited higher resistances, primarily due to lower thermal conductivity and higher contact resistance.

In the notebook platform shown in Fig. 2a, replacing the baseline TIM with FlexiMetal reduced steady-state junction temperature at a fixed processor power. As can be seen in Fig. 2b, the benefit was more pronounced under non-uniform die loading, where maintaining a relatively thin, high thermal conductivity continuous bond line across the die footprint is critical. Transient power step testing showed faster approach to steady-state and reduced incidence and magnitude of thermal and power throttling events relative to the baseline TIM.

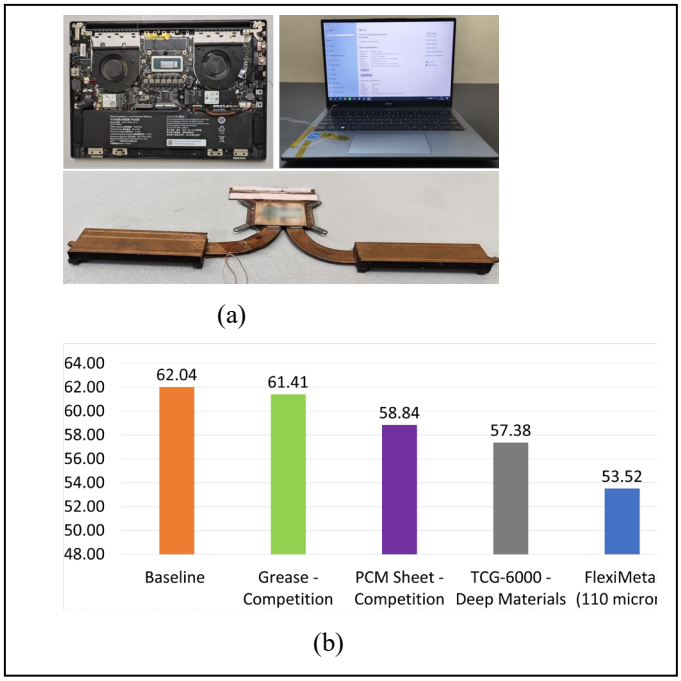


Fig. 2 a. Notebook platform and production thermal module employed for system-level TIM evaluation and (b) Test Data

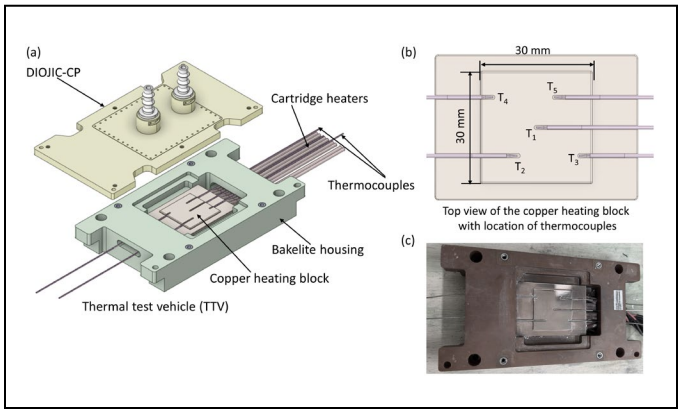


Fig. 3 Test section description: (a) schematic of the exploded views of the cold plate and TTV, (b) heater block with location of thermocouples, and (c) the thermal test vehicle (TTV)

In direct-to-chip liquid cooling at 1000 W TDP, FlexiMetal demonstrated significant thermal advantages over the reference grease TIM. All data below compare FlexiMetal and CryoFuze7 under identical flow conditions (0.75–2.5 L/min PG25, 30 × 30 mm² die, 45 °C inlet temperature) on the DIOJIC cold plate. Replacing the grease TIM with FlexiMetal reduced the center temperature by up to 6%, the maximum surface temperature by 6.5%, and the average surface temperature by 6.1% across the investigated flow range. More significantly, the thermal resistance (R_{th}) was reduced by up to 15% (peak values), maximum thermal resistance ($R_{th,max}$) by up to 10% and average thermal resistance ($R_{th,avg}$) by up to 11%, demonstrating FlexiMetal's superior interface conductance, as shown in Fig. 4. The thermal resistances discussed herein are calculated utilizing the temperature difference between the corresponding surface temperatures of the TTV and the fluid inlet temperature. Consequently, these thermal resistances include the thermal resistance of the TIM. At a nominal operating point of 2.0 L/min, the DIOJIC-CP with FlexiMetal achieved a maximum thermal resistance of approximately 0.032 °C/W, while attaining this performance at 60% lower pumping power compared to conventional skived-fin cold plates (CMC-CP) paired with FlexiMetal.

Temperature uniformity, quantified by the maximum absolute temperature difference (MATD) across the heated surface, improved with higher flow rates for both TIMs. At fixed flow, FlexiMetal enhanced the average heat transfer coefficient (h_{avg}) by up to 15% relative to grease, reflecting its superior conductivity and interfacial contact quality. This improvement was sustained across the tested flow range and corresponded to reduced spatial temperature gradients on the heated surface. See Fig. 4.

Notably, the choice between FlexiMetal and grease TIM had negligible impact on pressure drop and pumping power—differences were within measurement uncertainty (±1%) and dominated by the cold plate geometry rather than TIM properties. This decoupling confirms that FlexiMetal's benefits are purely thermal and mechanical (compliance, contact quality, pump-out suppression), with no hydraulic penalty [5]. These results validate that FlexiMetal is suitable not only for passive die-level cooling and confined spaces such as laptops, but also for next-generation liquid-cooled data center applications where soft metal TIM conductivity and stability directly reduce both die temperatures and system cooling power.

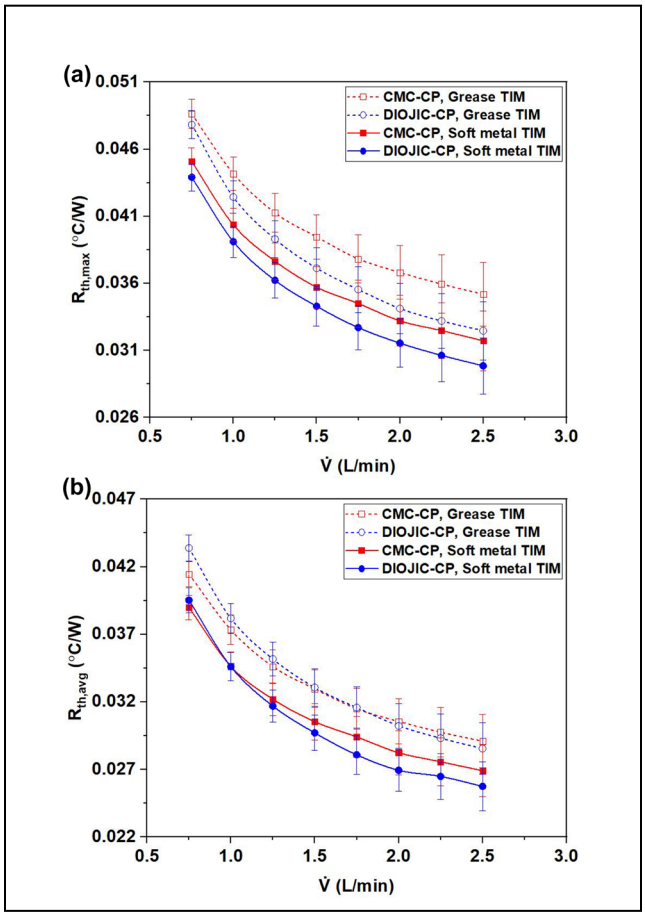


Fig. 4: Variation of thermal resistances: maximum thermal resistance ($R_{th,max}$), and average thermal resistance ($R_{th,avg}$) with volumetric flow rate ($\dot{V}$) for DIOJIC and CMC cold plates using two different TIMs.

V. Reliability Testing

A structured reliability test matrix was implemented at material, TTV, and system levels to evaluate the long-term stability of FlexiMetal. Passing criteria were defined as no significant directional drift in measured thermal performance beyond expected measurement scatter and no evidence of gross degradation mechanisms such as pump-out, cracking, or corrosive attack within the test window.

Thermal shock and environmental cycling included laptop assemblies and TTV-based configurations subjected to temperature excursions from approximately -40 °C to +65–75 °C. No measurable monotonic increase in interface resistance was observed, and inspections showed no pump-out, edge recession, or interfacial cracking.

High-temperature and high-humidity exposures at 85 °C and 85% relative humidity were conducted on bulk specimens and assembled copper/TTV configurations. Test duration was 1000 hours. FlexiMetal preserved its appearance and thermal performance, with no signs of corrosive interaction with copper within the tested durations.

High-temperature bake at 150 °C for durations up to and exceeding 1600 hours produced no observable phase separation, crust formation, or excessive bleed, and post-bake thermal performance remained consistent with pre-bake values. CPU and GPU power cycling on multiple notebook and gaming platforms over several thousand cycles showed stable junction temperatures and interface behavior, with no loss of interfacial wetting or emerging degradation trends.

Power cycling was performed in three notebook platforms to evaluate TIM stability and pump-out behavior under realistic thermo-mechanical loading. In each system, cycling drove the CPU temperature between approximately 40 °C and 100 °C (See Fig. 5a) inducing dynamic die curvature changes between nominally flat and convex states and thereby promoting classic pump-out mechanisms. Thermal performance was characterized at time zero and at periodic intervals; representative data from one notebook are shown below in Fig. 5b. The remaining systems exhibited similarly stable behavior, with negligible degradation even after almost 10 000 cycles. While the thermal shock, bake, and high humidity testing is a function of chemical properties of the material, power cycling and pump out is a physical property driven phenomenon.

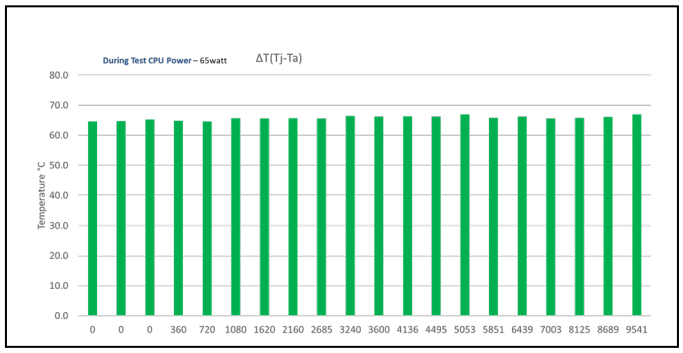


Fig. 5a: Temperature profile in the CPU during the power cycling to study pump out behavior

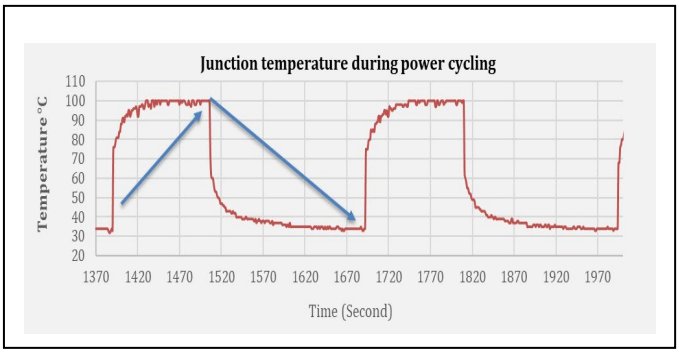


Fig. 5b: Steady state tests after thermal cycling showing almost no impact of almost 10,000 power cycles

VI. Rheological Characterization of Fleximetal and Impact on Pump Out

The rheological behavior of thermal interface materials plays a critical role in determining their long-term stability and resistance to pump-out under thermo-mechanical cycling. In this work, the viscoelastic properties of FlexiMetal were characterized using dynamic mechanical analysis, focusing on the storage shear modulus (G′) and loss shear modulus (G″), which respectively represent the elastic and viscous components of the material response.

Fig. 6 shows the measured G′ and G″ of FlexiMetal as a function of temperature. The data below about 80 °C are emphasized, as this temperature range corresponds to typical operating conditions of CPU and GPU thermal interface layers in most servers and client computing systems. Above this temperature range, the data exhibit increased noise, likely due to equipment stability issues.

Across the operating temperature window, FlexiMetal exhibits a storage modulus G′ in the range of approximately 0.2–0.6 MPa, while the loss modulus G″ remains significantly lower, in the range of approximately 0.05–0.15 MPa. The resulting ratio G′/G″ consistently exceeds unity, with typical values between 3 and 6. This indicates that, while FlexiMetal is compliant and conformal at the interface, its mechanical response is dominated by elastic energy storage rather than viscous flow.

This combination of high effective modulus and elastic-dominated response is unusual among thermal interface materials. Conventional thermal greases typically exhibit low G′ and $G'/G'' < 1$ in the operating temperature range, leading to flow-driven degradation mechanisms. In contrast, polymeric gel TIMs are specifically engineered to achieve $G'/G'' \geq 1$ to suppress pump-out, though they suffer from relatively higher thermal resistance due to higher thickness as well as higher contact resistances. FlexiMeta achieves similar or higher G′/G″ ratios than gels, while maintaining metallic thermal conductivity and ultra-low thermal resistance.

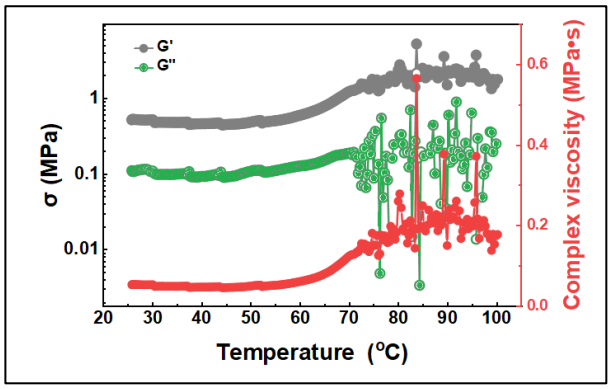


Fig. 6: Rheological data of FlexiMetal showing G' >G"

A. Pump-Out Resistance Theory and Application to FlexiMetal

Pump-out of thermal interface materials occurs when repeated thermo-mechanical cycling induces relative motion between the silicon die and the heat spreader or heat sink, causing the TIM to undergo shear deformation. If the TIM response is dominated by viscous flow, this deformation leads to progressive extrusion of the material from the interface, resulting in increased bond-line thickness, void formation, and degradation of thermal performance [2].

Prior work by Prasher and co-workers established a rheology-based framework for understanding pump-out resistance in thermal interface materials [1, 3]. In this framework, the resistance to pump-out is governed not by static viscosity alone, but by the relative contributions of elastic and viscous deformation, quantified by the ratio of storage to loss shear modulus, G'/G''. Materials with $G'/G'' \ll 1$ behave in a grease-like manner and are susceptible to pump-out, whereas materials with $G'/G'' \gg 1$ exhibit gel-like behavior and demonstrate strong resistance to pump-out under thermal cycling.

Prasher further demonstrated that both thermal contact resistance stability and degradation rate during temperature cycling correlate strongly with the G'/G'' ratio [2]. As G'/G'' increases, the degradation rate decreases sharply, indicating that elastic energy storage enables the TIM to accommodate cyclic shear strain without irreversible material transport. This relationship can be understood mechanistically as follows:

In a material dominated by viscous response (low G'/G''), each cycle of shear strain dissipates energy through irreversible flow. This viscous dissipation is accompanied by net material displacement—the material literally flows outward from the interface region under repeated cycling. Over thousands of cycles, this cumulative displacement results in thickening of the bond line, creation of voids, and progressive loss of thermal contact.

Conversely, in a material with elastic-dominated response (high G'/G''), shear strain is primarily accommodated through reversible elastic deformation. Energy is stored in elastic modes (e.g., distortion of the particle network) during compression and released upon unloading. Because the deformation is reversible, there is no net material transport per cycle. The material returns to its original position after each stress cycle, preventing the cumulative pump-out that degrades greases and low-modulus materials.

The transition between these regimes occurs near $G'/G'' \approx 1$, which physically corresponds to the gel point—the boundary between liquid-like (grease) and solid-like (gel) behavior. Materials that lack a gel point (formulations where G' never exceeds G'' even in the fully cured state) cannot develop the elastic-dominated response required to suppress pump-out, and they degrade in a manner identical to conventional greases [2].

B. Application to FlexiMetal

Applying this framework to FlexiMetal, the measured rheological data show that G'/G'' remains well above unity throughout the relevant operating temperature range. With G'/G'' values between approximately 3 and 6 below about 80 °C, FlexiMetal resides firmly within the high-reliability regime identified for gel-based TIMs, despite being an all-metallic material. Even above that temperature the ration of G'/G" is well above 1, but there is some experimental noise in the measurement data.

Physically, this behavior can be understood as follows. During power cycling, differential thermal expansion between the silicon die, package, and heat sink induces cyclic shear strain in the TIM layer. In FlexiMetal, this strain is primarily accommodated elastically, with energy stored and released during each cycle, rather than dissipated through viscous flow. As a result, there is minimal net material transport, preserving bond-line thickness and interfacial contact over extended cycling.

The quantitative relationship between G'/G'' and pump-out degradation can be expressed using Prasher's simplified framework :

$$R_c = c\left(\frac{\sigma}{G'/G''}\right)^n$$

where R_c is the contact resistance, σ is the surface roughness, G'/G'' is the ratio of storage to loss modulus, and c and n are empirical constants determined from experimental data. For structured TIMs with high G'/G'' ratios, contact resistance is minimized and remains stable under cycling, whereas materials lacking adequate G'/G'' experience substantial increases in contact resistance during thermal cycling due to void formation and material migration.

Within this framework, the long-term thermal resistance degradation rate, measured as the slope of resistance versus thermal cycle number, scales as:

$$\text{Degradation Rate} \propto (G'/G'')^{-m}$$

where m is an exponent typically in the range of 1–2. This relationship indicates that doubling the G'/G'' ratio can reduce the degradation rate by a factor of 2–4. For FlexiMetal, we measure G'/G'' values of 3–6 throughout the 40–70 °C range, substantially higher than both the early greases considered by Prasher and the commercial greases characterized by Nagrani and Marconnet. Within this simple scaling, such high G'/G'' values imply strongly suppressed contact-resistance growth. When combined with FlexiMetal's high intrinsic thermal conductivity arising from the metallic matrix, the material occupies a regime where both mechanical (voiding) and thermal degradation are minimized—analogous to the "high G', high k, high G'/η" (where η is the viscosity) design space identified as favorable for long-term thermal reliability in the recent accelerated-cycling studies [2].

This rheology-driven pump-out resistance provides a fundamental explanation for the observed system-level stability of FlexiMetal in prolonged power cycling and thermal stress tests. Unlike conventional low-viscosity liquid metal TIMs, which require physical dams or barriers to prevent leakage and extrusion, FlexiMetal inherently resists pump-out due to its elastic-dominated mechanical response. This intrinsic stability enables reliable long-term operation without additional mechanical containment structures, a significant advantage for manufacturing and field service.

The mechanism is further reinforced by the yield-stress character of FlexiMetal's gel structure. Above a critical shear stress threshold, the material can yield and flow locally to accommodate large surface asperities or non-ideal geometries during assembly. However, once assembly shear is removed, the elastic network recovers, and the material returns to a solid-

like state. This shear-responsive behavior allows the material to combine the conformability of greases (during assembly) with the cyclic-stability of rigid solids (during operation), without the pump-out vulnerability of low-modulus greases.

The complex viscoelastic response also imparts frequency-dependence to the material's behavior. At the high-frequency shear rates experienced during power cycling (driven by thermal transients on millisecond to second timescales), FlexiMetal behaves more elastically, further suppressing material transport. This frequency-dependent elasticity provides an additional layer of pump-out resistance beyond the static G'/G'' ratio alone.

VII. Conclusions

FlexiMetal is a metallic, gel-like thermal interface material developed to bridge the gap between high-conductivity metallic TIM concepts and the practical requirements of modern high-power electronics. The material provides low and repeatable thermal interface resistances at realistic assembly loads, maintains positional stability without pump-out or excessive migration, accommodates non-ideal flatness, large-area and multi-die geometries, and dynamic warpage, and withstands aggressive thermal shock, high-temperature, high-humidity, and power cycling exposures without significant directional drift in performance.

The combined TTV and system data highlight the importance of co-optimizing thermal conductivity, rheology, and interfacial mechanics in next-generation TIMs. FlexiMetal behaves as a structured metallic gel: at rest, its yield-stress and elastic character stabilize the bond line and reduce pump-out and long-term drift; under controlled shear during assembly, it can locally flow to displace air and conform to surface asperities and non-ideal topographies. After shear is removed, the structure recovers, helping preserve the improved contact geometry.

This behavior is advantageous for large dies, multi-die or chiplet-based modules, packages and lids with bow or warp, and cold plate assemblies where planarity deviations and dynamic warpage make it difficult to maintain uniformly thin interfaces using conventional greases or PCMs. Unlike rigid solder joints, FlexiMetal retains sufficient compliance to accommodate differential thermal expansion and mechanical cycling without cracking, while being substantially more stable than low-viscosity greases or liquid metals.

These attributes make FlexiMetal a strong candidate for high-performance client, data center CPU/GPU, accelerator, and cold-plate-cooled AI platforms where conventional TIM technologies are increasingly constrained. Future work will include expanded bond line thickness mapping, detailed rheological characterization across relevant temperatures and frequencies, and systematic evaluation of assembly process enhancements, such as controlled in-plane shear-conditioning, to further reduce thermal resistance and improve interface uniformity.

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